

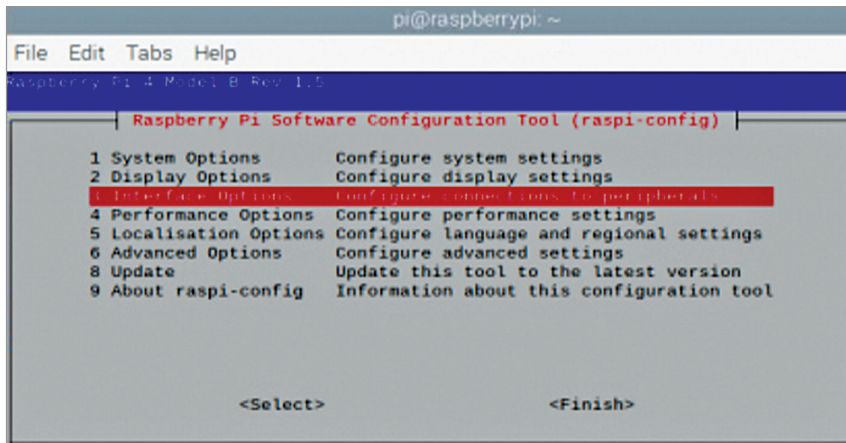


Fig. 3: Raspberry Pi configuration menu

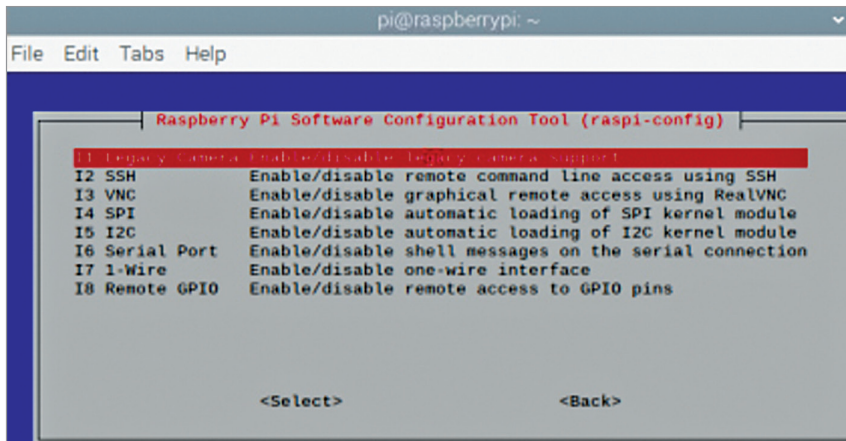


Fig. 4: Enabling camera configuration

and video storage with the Telegram bot. It is built around Raspberry Pi 4B+ (MOD1), a PIR motion sensor (S1), Pi Camera, and a few jumper wires.

Installation and setup

We have used Python for programming. Before coding, you need to configure Raspbian OS on the Raspberry Pi and install the necessary Pi camera library files.

To use a Raspberry Pi camera (connected via a ribbon cable), first enable the camera interface in the Raspberry Pi configuration settings. You can do this through the terminal using the following commands:

```
$ sudo apt-get install python-picamera
$ sudo apt-get install python3-picamera
```

You can also enable the camera interface using the Raspberry Pi configuration menu: Main Menu → Select Preferences → Raspberry Pi Configuration. Then, go to the Interfaces tab and enable the camera. Fig. 3 illustrates the Raspberry

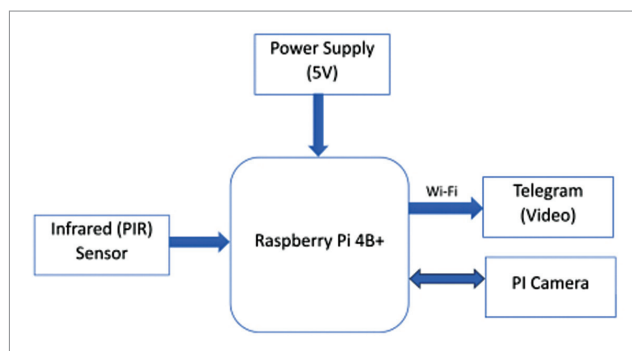


Fig. 5: Basic block diagram of the proposed setup

Pi configuration menu, and Fig. 4 shows the camera enabling configuration.

After successfully enabling the camera on the Raspberry Pi, it's recommended to perform software updates. This can be done by running the following Linux commands in the terminal:

```
$ sudo apt-get update
$ sudo apt-get upgrade
```

After updating and upgrading, reboot the system. Then, write the Python3/ThonnyIDE code for the bot. Fig. 5 presents a basic block diagram of the proposed setup.

Preparing Telegram bot

First, prepare a Telegram bot and extract the bot API. Follow the instructions for creating a Telegram bot using link: <https://core.telegram.org/bots/api>. To start using Telegram bots, issue the /start command to the BotFather, who will respond with a list of basic commands. To create a new bot:

1. Use the /newbot command.
2. Provide a name for your bot.
3. Choose a username for your bot.

If the names are available, you'll obtain the Token and the HTTP API key.

EFY note. This key is confidential; be careful about sharing it with anyone. Refer Fig. 6 for a clear understanding of the steps involved in getting the API. Note down the

API key to use in your code.

Coding

Next, install the Telegram bot Python module using 'sudo pip install telepot.' Note the following points:

1. For the proposed setup,